CMPG 325 COMPUTER NETWORKS

Individual Semester Project Final Submission: Technical Documentation and Evidence

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Assigned Organisation: Greater Taung Local Municipality Offices (Taung)

Industry : Municipal Services

1.EXECUTIVE SUMMARY

This report presents the design, implementation and testing of a computer network for Greater Taung Local Municipality Offices. The project focuses on providing reliable communication between internal devices, access to a central municipal server, dynamic routing between network segments and a separate guest Wi-Fi network for visitors.

The network was designed and implemented using Cisco Packet Tracer. The assigned address block, 172.30.70.0/23, was divided into smaller subnets for the internal LAN, VLAN 10, VLAN 20 and the point-to-point connection between the two routers.

The implementation consists of two routers, two switches, a municipal server, internal workstations, a wireless access point and a guest client. VLANs were configured to separate internal and guest traffic, while OSPF was configured in Area 0 to allow the routers to learn routes dynamically.

The main networking challenge was to implement Basic OSPF Single-Area Dynamic Routing. The additional change request required guest Wi-Fi users to be prevented from accessing internal municipal resources. An access control list (ACL) was therefore configured on the guest VLAN sub interface of R2 to block traffic from the guest subnet to the internal networks.

Connectivity testing confirmed that internal devices could reach the municipal server and communicate across the router connection. The guest client could reach its default gateway but could not reach the internal HR workstation or municipal server after the ACL was applied. These results demonstrated that the main routing and guest-isolation requirements had been implemented successfully.

2. INTRODUCTION

Computer networks are important in municipal offices because employees need to communicate, share resources and access services required for daily operations. A properly designed network should provide reliable connectivity while controlling access to internal resources.

This project involved designing and implementing a network for Greater Taung Local Municipality Offices using Cisco Packet Tracer. The project was based on the requirements provided in the CMPG 325 individual semester project.

The assigned network address block was 172.30.70.0/23. The assigned networking challenge was Basic OSPF Single-Area Dynamic Routing, and Change Request 3 required the implementation of guest Wi-Fi that is isolated from internal municipal resources.

The project progressed from an initial design review to the implementation and testing of the network. The final implementation included subnetting, VLAN configuration, router subinterfaces, trunking, OSPF routing and an access control list to restrict guest access.

The purpose of this report is to explain the network requirements, design decisions, IP addressing plan, implementation, verification, troubleshooting and final results.

3. PROJECT OBJECTIVES

The main objective was to design and implement a working municipal network that meets the client's connectivity and guest-access requirements.

The specific objectives were:

- To design a network topology suitable for the municipal office environment.
- To divide the assigned 172.30.70.0/23 address block into smaller subnets.
- To configure two routers and two switches to connect the network devices.
- To configure VLAN 10 for internal devices and VLAN 20 for guest devices.
- To configure the switch trunk and router subinterfaces required for VLAN communication.
- To implement OSPF in a single area so that the routers can learn routes dynamically.
- To provide connectivity to the municipal server from internal workstations.
- To implement guest Wi-Fi while restricting guest access to internal municipal resources.
- To test connectivity and verify the operation of the routing, VLAN and access-control configurations.
- To identify and resolve problems encountered during implementation.

4. CLIENT REQUIREMENTS

The network was designed according to the requirements of Greater Taung Local Municipality Offices.

4.1 Reliable Network Connectivity

The network must allow devices in different parts of the municipal office environment to communicate. Internal workstations must be able to reach the appropriate network devices and services.

4.2 Internal Resource Access

Municipal employees must be able to communicate with other internal devices and access the central municipal server. The server represents the file, printing and application services required by the client.

4.3 Guest Wi-Fi

Visitors must be able to connect to a separate guest network through a wireless access point. Guest users should not be given unrestricted access to internal municipal resources.

4.4 Network Segmentation

The network must separate internal devices from guest devices. VLANs provide logical separation at the switching level, while routing and access-control rules determine which traffic can pass between network segments.

4.5 Dynamic Routing

The project requires Basic OSPF Single-Area Dynamic Routing. OSPF must allow the two routers to exchange routing information and learn routes to the relevant internal networks.

4.6 Availability of Critical Services

The network design must support connectivity to the server representing critical municipal services. The network should allow internal users on both sides of the router connection to reach this server.

4.7 Testing and Verification

The implementation must be testable in Cisco Packet Tracer. Connectivity tests and configuration commands are used to verify routing, VLANs, trunking and guest-network restrictions.

5. NETWORK DESIGN AND TOPOLOGY

5.1 Overview of the Network Design

The network was designed using two routers, two switches, a municipal server, internal workstations, a wireless access point and a guest client.

R1 serves as the Core Router, connecting the internal LAN to the rest of the network. R2 serves as the Branch Router and provides connectivity to VLAN 10 and VLAN 20 through router subinterfaces.

The two routers are connected through a point-to-point link. This link provides communication between routers and allows OSPF routing information to be exchanged.

SW1 connects the internal devices on the R1 side. SW2 connects devices assigned to the different VLANs. A trunk connection between SW2 and R2 carries the VLAN traffic required for routing through the router subinterfaces.

The municipal server is connected to the internal LAN. Admin-PC and Finance-PC are also on this LAN. HR-PC and Services-PC are assigned to VLAN 10, while the guest client connects through the wireless access point and belongs to VLAN 20.

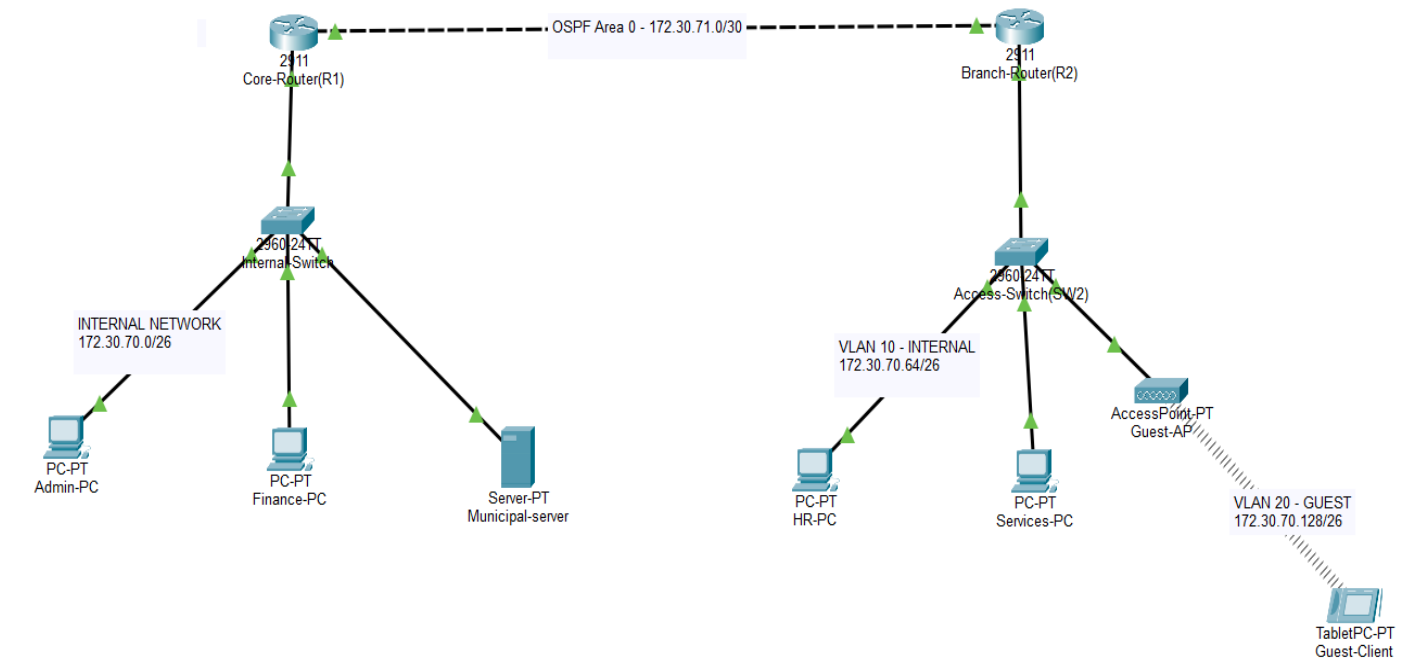


Figure 1: Cisco Packet Tracer network design

5.2 Network Devices and Their Roles

- **R1 – Core Router:**

R1 connects the internal LAN to the point-to-point link leading to R2. It provides the default gateway for the internal LAN and participates in OSPF routing.

- **R2 – Branch Router:**

R2 connects to R1 and provides routing for VLAN 10 and VLAN 20 through its subinterfaces. It participates in OSPF and applies the access control list used to restrict guest traffic.

- **SW1 – Internal Switch:**

SW1 provides wired connectivity for the internal LAN devices, including Admin-PC, Finance-PC and the municipal server.

- **SW2 – Access Switch:**

SW2 connects the devices assigned to VLAN 10 and VLAN 20. It provides access ports for the relevant devices and a trunk connection carrying VLAN traffic to R2.

- **Municipal-Server:**

The municipal server represents the central server hosting the file, printing and application services required by the client. Its IP connectivity was tested from internal workstations.

- **Admin-PC and Finance-PC:**

These workstations represent internal municipal users connected to the LAN on the R1 side.

- **HR-PC and Services-PC:**

These workstations belong to VLAN 10, named INTERNAL. They represent additional internal devices connected through SW2 and R2.

- **Wireless Access Point and Guest-Client:**

The wireless access point provides a connection for the guest client. The guest client belongs to VLAN 20, named GUEST, which is restricted from accessing the internal networks by the configured ACL.

5.3 Physical and Logical Organisation

The physical topology represents the connections between the routers, switches, server, workstations and wireless access point. The logical topology represents how devices are grouped into network segments and how traffic is routed between them.

The implementation uses VLANs to separate internal and guest traffic logically. Router subinterfaces allow R2 to provide gateway services for VLAN 10 and VLAN 20 over the trunk connection.

The point-to-point link between R1 and R2 provides the connection between the internal LAN and the networks connected to R2.

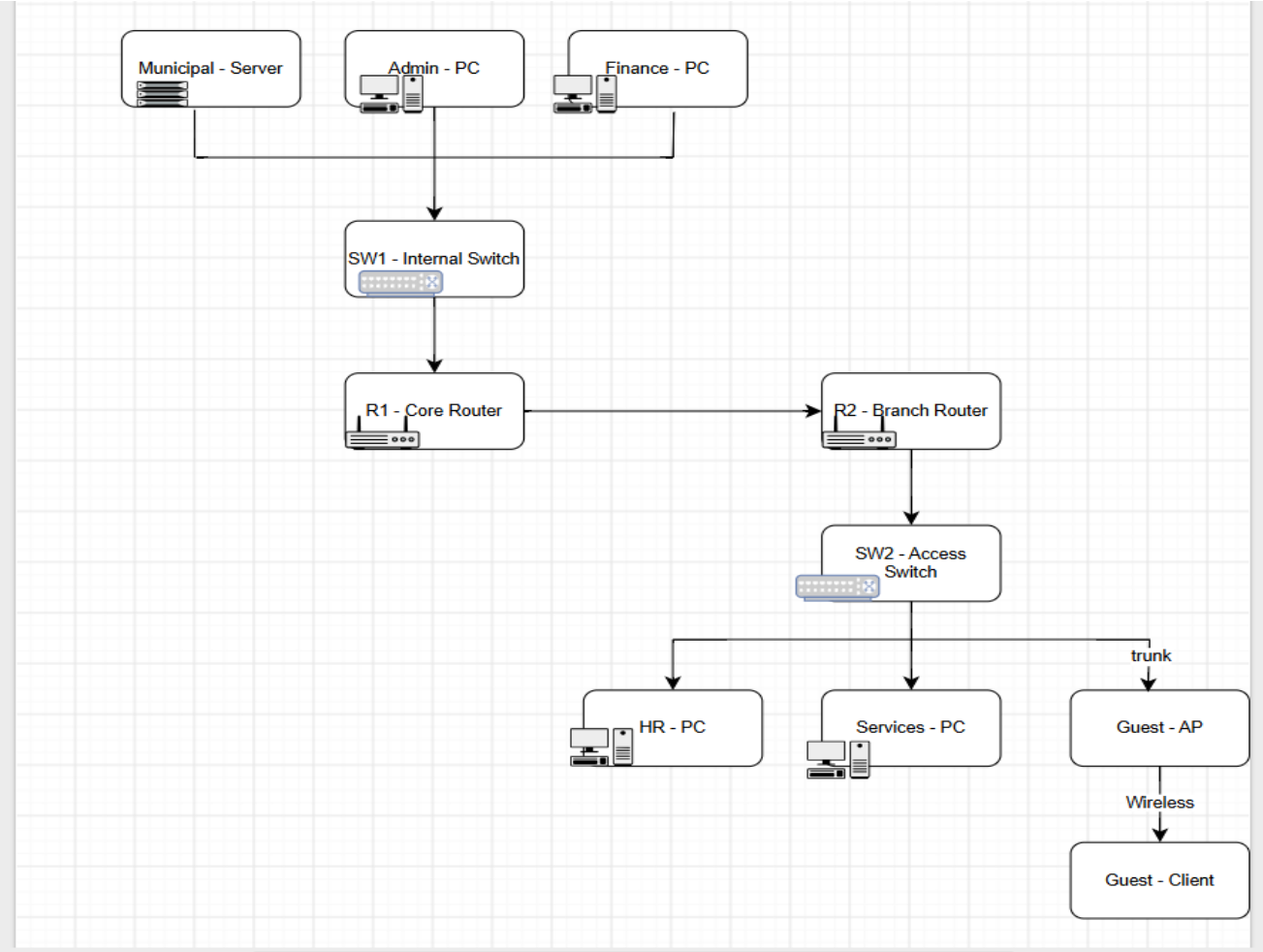


Figure 2: Physical Topology (draw.io)

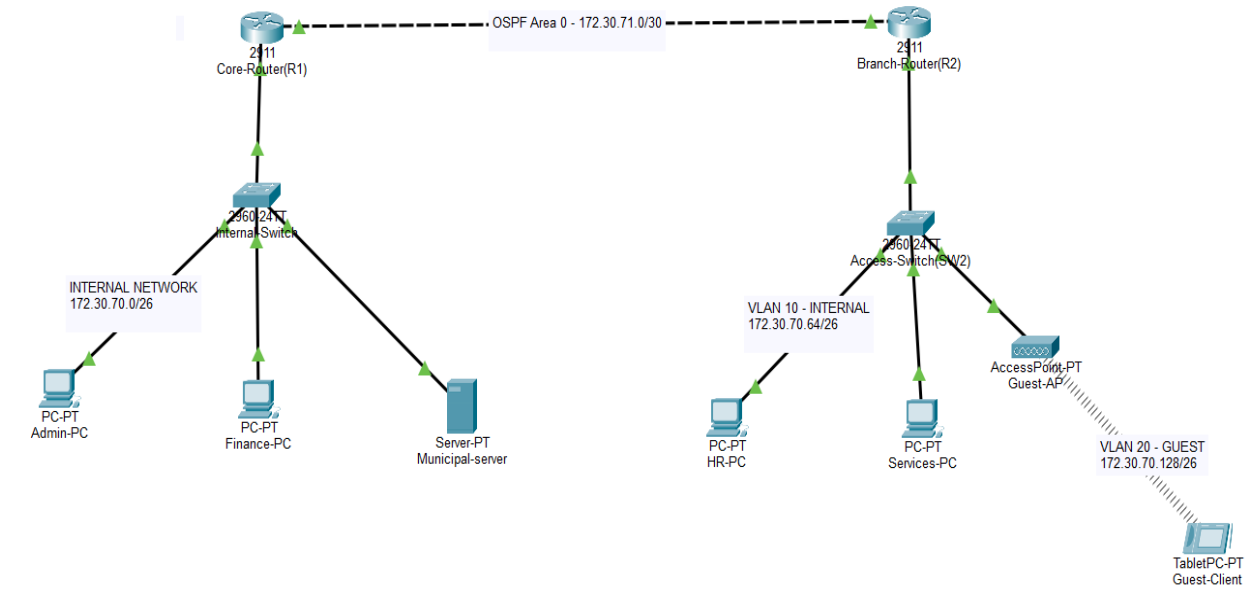


Figure 3: logical Topology

5.4 Design Decisions

Two routers were used to represent the core and branch portions of the network. This arrangement provides an opportunity to implement and demonstrate dynamic routing between different network segments.

VLANs were used to separate internal and guest devices. This allows the network to maintain distinct logical groups even when devices connect through the same access switch.

A trunk link was used to carry VLAN traffic between SW2 and R2. Router subinterfaces were configured to provide gateway addresses for the two VLANs.

OSPF was selected because it supports dynamic route learning between routers. This reduces the need to configure every remote network route manually.

An access control list was added to address the guest-isolation requirement. VLAN separation and OSPF configuration alone were not sufficient to enforce the required restriction. The ACL blocks guest traffic destined for the internal address ranges while permitting other IP traffic.

6. IP ADDRESSING PLAN

The network uses the assigned private IPv4 address block, 172.30.70.0/23. This address block was divided into smaller subnets to accommodate the internal LAN, VLAN 10 (INTERNAL), VLAN 20 (GUEST), and the point-to-point link between the two routers. The tables below present the subnet allocation, router interface addresses, and end-device IP configurations.

Network Segment	**Network Address**	**Subnet Mask**	**Default Gateway**
Core-Router Internal LAN	172.30.70.0/26	255.255.255.192	172.30.70.1
Internal VLAN 10	172.30.70.64/26	255.255.255.192	172.30.70.65
Guest VLAN 20	172.30.70.128/26	255.255.255.192	172.30.70.129
Core-Router–R2 Link	172.30.71.0/30	255.255.255.252	—

Table 1:

The router interfaces are configured as follows:

Device	**Interface**	**IP Address**
Core-Router (R1)	GigabitEthernet0/1	172.30.70.1
Core-Router (R1)	GigabitEthernet0/0	172.30.71.1
R2	GigabitEthernet0/0	172.30.71.2
R2	GigabitEthernet0/1.10	172.30.70.65
R2	GigabitEthernet0/1.20	172.30.70.129

Table 2: router interfaces

Individual devices on the network are addressed as follows:

Device	IP Address	Subnet Mask	Default Gateway
Admin-PC	172.30.70.11	255.255.255.192	172.30.70.1
Finance-PC	172.30.70.12	255.255.255.192	172.30.70.1
Municipal-Server	172.30.70.10	255.255.255.192	172.30.70.1
HR-PC (VLAN 10)	172.30.70.66	255.255.255.192	172.30.70.65
Services-PC (VLAN 10)	172.30.70.67	255.255.255.192	172.30.70.65
Guest-Client (VLAN 20)	172.30.70.130	255.255.255.192	172.30.70.129

Table 3: individual devices

7. ASSIGNED FEATURE IMPLEMENTATION

7.1 Implementation Environment

The network was implemented and tested using Cisco Packet Tracer. The simulation environment allowed the routers, switches, server, workstations and wireless access point to be configured and tested without requiring physical networking equipment.

The implementation followed the network design and addressing plan established during the earlier design stage.

7.2 BASIC OSPF SINGLE AREA DYNAMIC ROUTING

The assigned networking challenge was implemented using Open Shortest Path First (OSPF) between the Core Router (R1) and Branch Router (R2).

OSPF was configured using Area 0. The connection between R1 and R2 uses the network 172.30.71.0/30.

R1 advertises the internal LAN network 172.30.70.0/26, while R2 advertises the internal VLAN 10 network 172.30.70.64/26.

The guest VLAN 20 network, 172.30.70.128/26, was not included in the OSPF network statements. Guest isolation is handled separately using an ACL on R2.

OSPF Neighbour Verification

The OSPF neighbour relationship was checked on both routers using the show **ip ospf neighbor command.**

R1 shows R2 in the FULL/DR state, and R2 shows R1 in the FULL/BDR state. This confirms that the OSPF neighbour relationship was successfully established.

```
Router>enable
Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
172.30.71.1       1   FULL/BDR        00:00:35    172.30.71.1     GigabitEthernet0/0
Router#
```

Figure 4: R2 OSPF neighbour verification

```
Router>enable
Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
172.30.71.2       1   FULL/DR         00:00:34    172.30.71.2     GigabitEthernet0/0
Router#
```

Figure 5: R1 OSPF neighbour verification

OSPF Routing Verification

The routing table on R2 was checked using **show ip route**.

The output shows:

O 172.30.70.0/26 [110/2] via 172.30.71.1

The O indicates that the route was learned through OSPF. This confirms that R2 is dynamically learning the R1 internal LAN network.

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.30.0.0/16 is variably subnetted, 7 subnets, 3 masks
O       172.30.70.0/26 [110/2] via 172.30.71.1, 00:29:10, GigabitEthernet0/0
C       172.30.70.64/26 is directly connected, GigabitEthernet0/1.10
L       172.30.70.65/32 is directly connected, GigabitEthernet0/1.10
C       172.30.70.128/26 is directly connected, GigabitEthernet0/1.20
L       172.30.70.129/32 is directly connected, GigabitEthernet0/1.20
C       172.30.71.0/30 is directly connected, GigabitEthernet0/0
L       172.30.71.2/32 is directly connected, GigabitEthernet0/0

Router#
```

Figure 6: R2 routing table showing the OSPF-learned route

7.3 GUEST WIFI AND ISOLATION

Change Request 3 was implemented by creating a separate Guest VLAN 20 for visitors. VLAN 20 uses the network 172.30.70.128/26, with 172.30.70.129 configured as the gateway. The Guest-Client uses 172.30.70.130.

The guest network is connected through the wireless access point and VLAN 20 is carried across the switch trunk.

VLAN Configuration

The VLAN configuration was verified using **show vlan brief.**

The output confirms that VLAN 10, named INTERNAL, and VLAN 20, named GUEST, are active.

```
Switch>enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   INTERNAL                         active    Fa0/2, Fa0/3
20   GUEST                            active    Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Figure 7: Access Switch (SW2) VLAN configuration

Trunk Configuration

The connection between SW2 and R2 was verified using **show interfaces trunk**.

The output confirms that the trunk uses 802.1Q and that VLANs 10 and 20 are active and forwarding.

```
1005 trnet-default                    active
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20

Switch#
```

Figure 8: Access-Switch (SW2) trunk configuration

GUEST ISOLATION ACL

An extended ACL was configured on R2's G0/1.20 sub interface to prevent guest traffic from reaching the internal networks.

The ACL blocks traffic from 172.30.70.128/26 to the internal range 172.30.70.0/25 while allowing other traffic.

The ACL was verified using **show access-lists 100**. The deny statement recorded matches, confirming that guest-to-internal traffic was being blocked.

```
Router#show access-lists 100
Extended IP access list 100
    deny ip 172.30.70.128 0.0.0.63 172.30.70.0 0.0.0.127 (8 match(es))
    permit ip any any (4 match(es))

Router#
```

Copy

Figure 9: ACL 100 verification

8. TESTING AND VERIFICATION

Testing was performed to confirm that the network was functioning correctly and that the implemented features behaved as intended.

8.1. INTERNAL NETWORK CONNECTIVITY

Admin-PC to HR-PC

A ping was performed from Admin-PC to HR-PC at 172.30.70.66.

The test returned 4/4 successful replies with 0% packet loss, confirming connectivity between the internal LAN and VLAN 10.

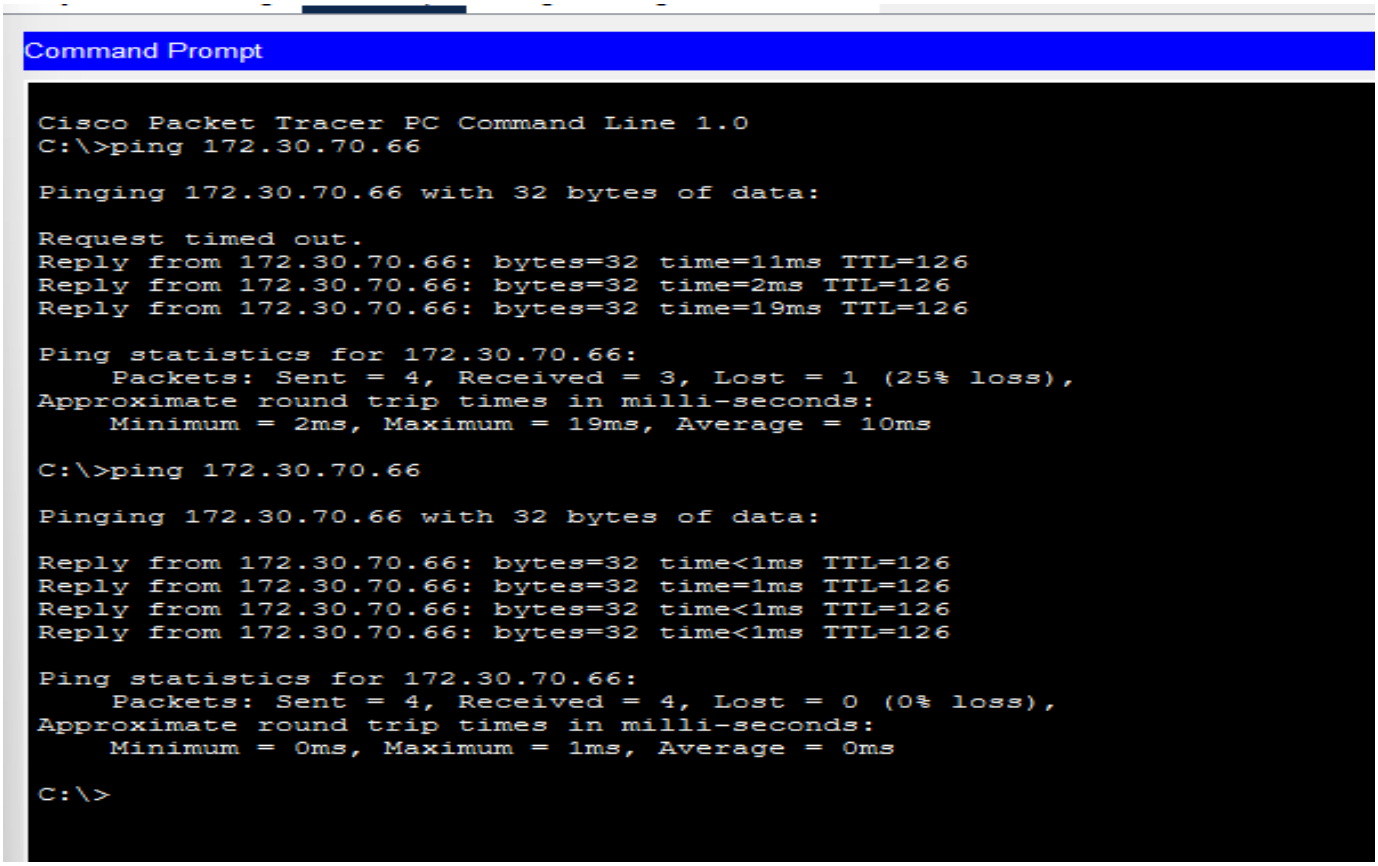


Figure 10: Admin-PC to HR-PC connectivity test

Admin-PC to Municipal-Server

A ping was performed from Admin-PC to the Municipal-Server at 172.30.70.10.

The test returned 4/4 successful replies with 0% packet loss, confirming network-level connectivity between the internal workstation and the municipal server.

The Municipal-Server represents the server used for the municipal file, print and application services required by the project. This test confirms network connectivity to the server.

```
C:\>ping 172.30.70.10

Pinging 172.30.70.10 with 32 bytes of data:

Reply from 172.30.70.10: bytes=32 time<1ms TTL=128
Reply from 172.30.70.10: bytes=32 time<1ms TTL=128
Reply from 172.30.70.10: bytes=32 time<1ms TTL=128
Reply from 172.30.70.10: bytes=32 time<1ms TTL=128

Ping statistics for 172.30.70.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 11: Admin-PC to Municipal-Server connectivity test

HR-PC to Municipal-Server

The project requires the critical municipal services to remain available during business hours. To test connectivity to the server from another part of the network, a ping was performed from HR-PC to the Municipal-Server at 172.30.70.10.

The test returned 4/4 successful replies with 0% packet loss, confirming that the Municipal-Server is reachable from the R2-side internal network through the OSPF connection. This demonstrates that users on this side of the network can reach the server hosting the critical municipal services.

```
C:\>ping 172.30.70.10

Pinging 172.30.70.10 with 32 bytes of data:

Reply from 172.30.70.10: bytes=32 time<1ms TTL=126
Reply from 172.30.70.10: bytes=32 time<1ms TTL=126
Reply from 172.30.70.10: bytes=32 time=11ms TTL=126
Reply from 172.30.70.10: bytes=32 time<1ms TTL=126

Ping statistics for 172.30.70.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 2ms

C:\>
```

Figure 12: HR-PC to Municipal-Server connectivity test

8.2 GUEST NETWORK CONNECTIVITY

Guest-Client to Guest Gateway

The Guest-Client was tested against its gateway at 172.30.70.129.

The test returned 4/4 successful replies with 0% packet loss, confirming that the Guest-Client had working connectivity within the guest network.

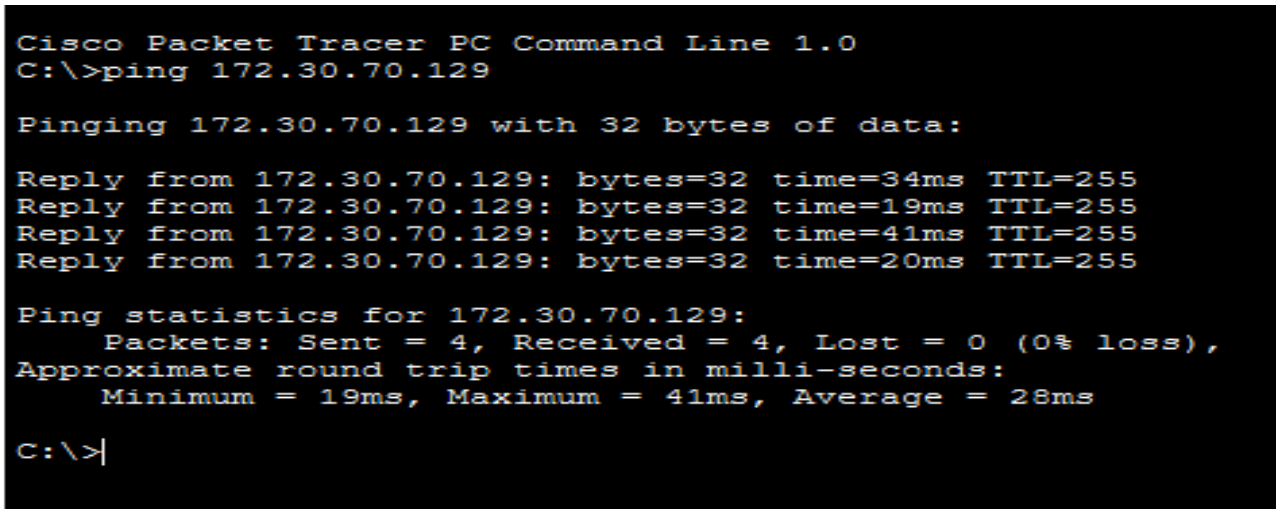


Figure 13: Guest-Client to guest gateway connectivity test

8.3 GUEST NETWORK ISOLATION

Guest network isolation was tested by attempting to reach internal municipal resources from the Guest-Client.

Guest-Client to HR-PC

Before the ACL was implemented, the Guest-Client was able to reach HR-PC. This showed that the guest network was connected but was not yet isolated from the internal network.

After ACL 100 was applied, the test was repeated.

The Guest-Client was unable to reach HR-PC, resulting in 0/4 successful replies and 100% packet loss. The response showed: Reply from 172.30.70.129: Destination host unreachable.

This confirms that guest-to-internal traffic was being blocked.

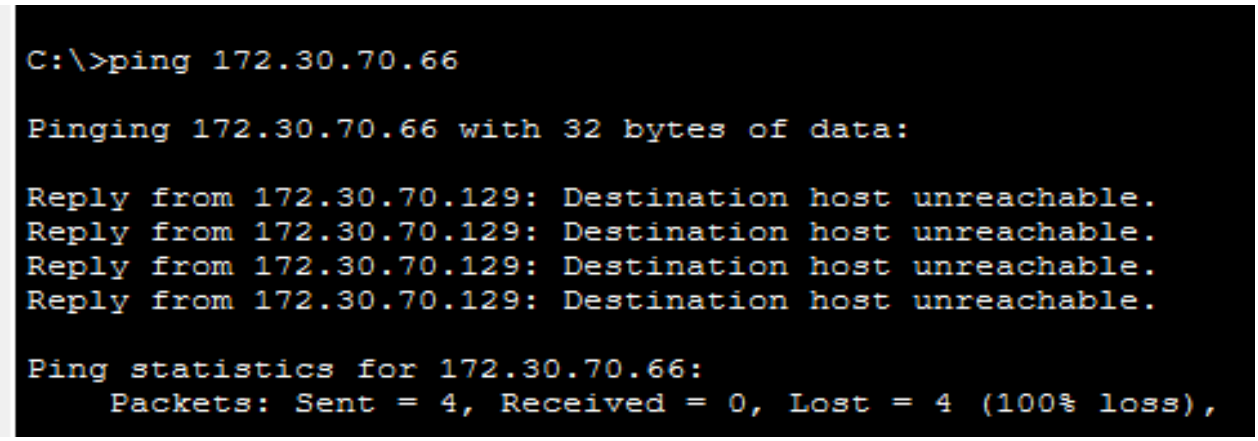


Figure 14: Guest-Client to HR-PC isolation test

Guest-Client to Municipal-Server

The Guest-Client was also tested against the Municipal-Server at 172.30.70.10.

The test resulted in 0/4 successful replies and 100% packet loss, with the guest gateway returning a destination unreachable message. This confirms that the guest network cannot reach the municipal server.

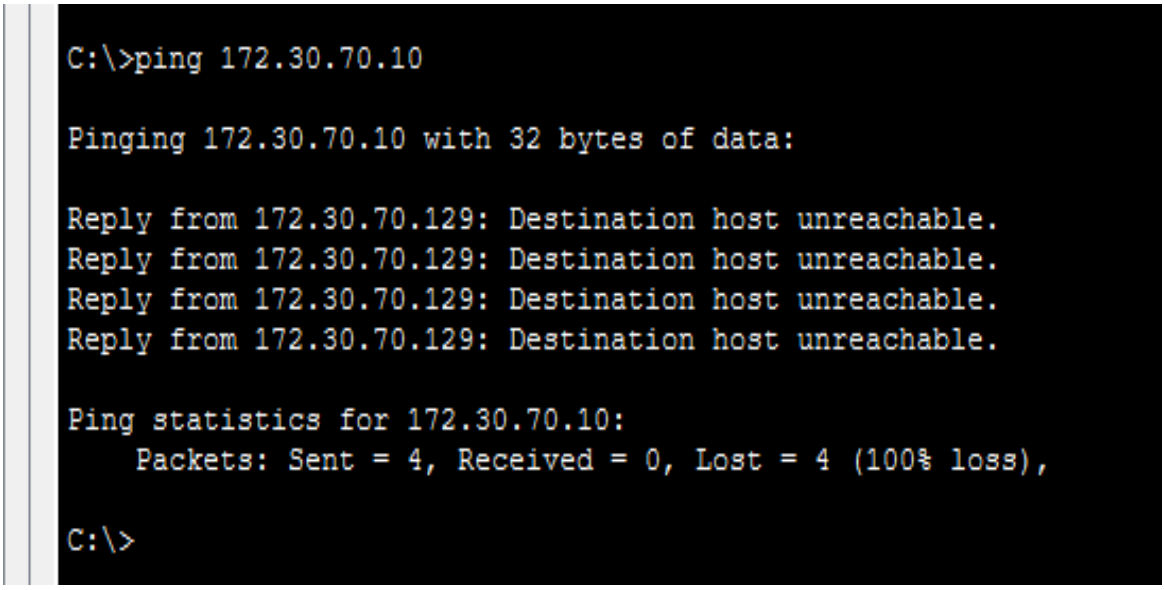


Figure 15: Guest-Client to Municipal-Server isolation test

9. TROUBLESHOOTING

During implementation, testing was used to identify problems and confirm that the final network behaviour matched the project requirements.

9.1. GUEST WIRELESS DEVICE

The original Guest-Client was configured as a Laptop-PT. The device did not have a Wireless0 interface available in the Packet Tracer version being used, and the wireless interface could not be installed.

The issue was resolved by replacing the Laptop-PT with a TabletPC-PT, which supported wireless connectivity.

9.2 GUEST-CLIENT CONFIGURATION

The replacement Guest-Client was configured with:

- IP address: 172.30.70.130
- Subnet mask: 255.255.255.192
- Default gateway: 172.30.70.129

The Guest-Client was then tested against its gateway and successfully received replies.

This is shown in Figure 13.

9.3. GUEST ISOLATION

After the Guest-Client was successfully connected, a test to HR-PC initially succeeded. This showed that the guest network was working but guest traffic could still reach the internal network.

An ACL was therefore configured on R2's G0/1.20 subinterface to block traffic from the guest subnet to the internal networks.

The test was repeated after applying the ACL, and guest access to HR-PC and the Municipal-Server was successfully blocked.

The ACL was also checked using show access-lists 100, which showed matches on the deny statement.

This is shown in Figure 9, Figure 14, Figure 15.

10. TESTING SUMMARY

Test/Verification	Result	Evidence
R2 OSPF neighbour	FULL/BDR	Figure 4
R1 OSPF neighbour	FULL/DR	Figure 5
OSPF route on R2	R1 LAN learned through OSPF	Figure 6
Admin-PC to HR-PC	4/4, 0% loss	Figure 10
Admin-PC to Municipal-Server	4/4, 0% loss	Figure 11
HR-PC to Municipal-Server	4/4, 0% loss	Figure 12
Guest-Client to Gateway	4/4, 0% loss	Figure 13
Guest-Client to HR-PC	0/4, 100% loss	Figure 14
Guest-Client to Municipal-Server	0/4, 100% loss	Figure 15
ACL 100	Deny rule recorded matches	Figure 9
VLAN configuration	VLANs 10 and 20 active	Figure 7
Trunk configuration	VLANs 10 and 20 forwarding	Figure 8

11. PROJECT OUTCOMES

The project resulted in a working Packet Tracer network that implements the main design and routing requirements.

The assigned /23 address block was divided into smaller subnets for the internal LAN, VLAN 10, VLAN 20 and the point-to-point link.

The two routers were configured with the required interface addresses and routing configuration. The OSPF neighbour relationship was established, and R2 learned the internal LAN route through OSPF.

VLAN 10 and VLAN 20 were configured to separate internal and guest devices. The switch trunk and router subinterfaces provided the required VLAN connectivity.

Internal connectivity tests succeeded, including tests from Admin-PC to HR-PC, Admin-PC to the municipal server and HR-PC to the municipal server.

Guest isolation was implemented with an ACL. The guest client could reach its gateway, but its attempts to reach HR-PC and the municipal server were blocked after the ACL was applied.

The final Packet Tracer file, milestone documents, README and supporting evidence were organised in the project's GitHub repository(https://github.com/Phathu647/CMPG325-Greater-Taung-Municipal-Network.git)

12. LIMITATIONS AND POSSIBLE FUTURE IMPROVEMENTS

The project demonstrates the required routing, VLAN segmentation, IP addressing and guest-access restrictions in a simulated network environment.

The implementation was tested using IP connectivity and configuration verification commands. The successful ping tests confirm network-layer reachability to the municipal server, but they do not independently verify the operation of specific file-sharing, printing or municipal application services.

The project could be extended in the future by configuring and testing the relevant application services on the server, adding further security controls and testing additional failure scenarios.

In a production municipal environment, further work could also include redundant network links, backup equipment, monitoring, documented recovery procedures and more detailed wireless security configuration. These are possible future improvements rather than features claimed as implemented in this project.

13. REFLECTION

This project provided practical experience in designing, configuring and testing a network using Cisco Packet Tracer.

One of the main learning outcomes was understanding how subnetting, VLANs, router subinterfaces and dynamic routing work together. Dividing the assigned address block into smaller subnets helped organise the network and made it easier to identify the devices and gateways belonging to each segment.

The OSPF implementation provided experience in establishing a routing neighbour relationship and verifying that a router had learned a remote network dynamically. The routing table and OSPF neighbour commands were particularly useful for confirming that the routers were communicating correctly.

The guest-isolation requirement was another important learning experience. The initial test showed that the guest client could reach an internal workstation even though the guest subnet was excluded from OSPF advertisements. This helped demonstrate that routing advertisements and traffic filtering serve different purposes. Applying the ACL and repeating the tests provided evidence that the guest-to-internal restriction was working.

Overall, the project improved my understanding of network design, configuration, verification and troubleshooting. It also demonstrated the importance of testing a network against the client's actual requirements rather than relying only on the configuration commands.

14. CONCLUSION

This report presented the design, implementation and testing of the network for Greater Taung Local Municipality Offices.

The network was implemented in Cisco Packet Tracer using two routers, two switches, a municipal server, internal workstations, a wireless access point and a guest client. The assigned 172.30.70.0/23 address block was divided into subnets for the internal LAN, VLAN 10, VLAN 20 and the point-to-point router connection.

OSPF was configured in Area 0, and verification confirmed that the routers formed a FULL neighbour relationship and that R2 learned the internal LAN route through OSPF. VLAN 10 and VLAN 20 were configured, and the trunk connection was verified.

The guest-isolation requirement was addressed using ACL 100 on R2. Testing confirmed that the guest client could reach its default gateway but could not reach the tested internal workstation or municipal server after the ACL was applied.

The project met its main simulated network connectivity, routing and guest-isolation objectives. The final Packet Tracer implementation and supporting evidence provide a basis for reviewing the configuration and the test results.

15. GITHUB PORTFORLIO

https://github.com/Phathu647/CMPG325-Greater-Taung-Municipal-Network.git